



Amlogic Widevine Provision 4.0

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Change History

Revision	Revised Date	By	Changes
1.0	2023/08/08	shuo.zhang	Initial release draft

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Audience

Provisioning 4.0 is the next generation of Widevine device provisioning. This provisioning scheme is closely aligned with a model designed by the Trusted Computing Group. By intent and design, the Widevine Provisioning 4.0 model is closely aligned with Android's new Remote Attestation Key Provisioning process. This model is based on using the secure boot process to build a certificate chain. In this model, each device is seeded with a unique device secret value referred to as UDS.

This document is intended OEM device manufacturers who use Amlogic Chipset integrate with Widevine L1 Provisioning 4.0.

1. Device Model Register

If OEM hasn't registered model in widevine integration console, please contact widevine team and register a model in widevine integration console. OEM needs to provide chipset serial and max hdcv support level of device to FAE to acquire SOC Security Profiles from Amlogic. SOC Security Profiles information will be used when register new device model.

OEM can register widevine client by <https://bit.ly/widevine-newclient>.

2. Extraction Widevine CSR

2.1 Android

2.1.1 Extra by rkp_factory_extraction_tool

For Android ATV, google has implemented rkp_factory_extraction_tool to acquire Remote Provision CSR and widevine CSR. OEM can also refer to Remote Key Provisioning for more information.

The source code of extraction tool is located in the AOSP source tree.

```
system/security/provisioner/rkp_factory_extraction_tool.cpp
$ m rkp_factory_extraction_tool
$ adb root
$ adb push
\out/target/product/<<product_name>>/vendor/bin/rkp_factory_extraction_tool
\data
$ adb shell chmod 777 /data/rkp_factory_extraction_tool
$ adb shell export LD_LIBRARY_PATH=/vendor/lib:/vendor/lib64:$LD_LIBRARY_PATH
$ adb shell /data/rkp_factory_extraction_tool > csr.json --output_format
build+csr
```

For example one csr.json which contains RPK and Widevine CSR.

```
{"build_fingerprint": "****", "csr": "hQGggqQBAQMnIAYhwCA****", "name": "default", "serialno": "ohm8096a104c1010319"}
{"build_fingerprint": "****", "csr": "hQGggqQBAQMnIAYhwCA****", "name": "widevine", "serialno": "ohm8096a104c1010319"}
```

2.1.2 Extra by `wv_factory_extraction_tool`

OEM also uses `wv_factory_extraction_tool` to get Widevine CSR.

The source code of `wv_factory_extraction_tool` in Widevine Cdm:

```
vendor/widevine/libwvdrmengine/tools/factory_upload_tool/

$ mmm vendor/widevine/libwvdrmengine/tools/factory_upload_tool/
$ adb push
\out/target/product/<<product_name>>/vendor/bin/hw/wv_factory_extraction_tool
\data
$ adb shell chmod 777 /data/wv_factory_extraction_tool
$ fingerprint=`adb shell getprop ro.build.fingerprint`
$ serialno=`adb shell getprop ro.serialno`
$ model=`adb shell getprop ro.product.model`
$ csr_v3=`adb shell "export
LD_LIBRARY_PATH=/vendor/lib:/vendor/lib64:$LD_LIBRARY_PATH &&
/data/wv_factory_extraction_tool csr_v3 | base64 -w 0"
$echo "{\"build_fingerprint\":\"$fingerprint\",\"csr\":\"$echo
$csr_v3\",\"name\":\"widevine\",\"serialno\":\"$serialno\"}" > $
model_$serialno.json
$ cat $model_$serialno.json
```

For example, one device json:

```
{"build_fingerprint": "****", "csr": "hQGggqQBAQMnIAYhwCA****", "name": "widevine", "serialno": "ohm8096a104c1010319"}
```

2.1.3 Extra by OEM Apk or Native App

Notice:

OEMS can customize the way to obtain CSR according to their own needs. However, please ensure that the format of chr csr is same with Extra by `rkp_factory_extraction_tool`.

3. Upload Widevine CSR

1. Set Up Android Partner API

OEM should register a company account of GCP (Google Cloud Project). Please refer to <https://docs.partner.android.com/gms/building/integrating/att-keys/Setting-Up-Partner-API>.

2.Upload CSR Request

OEM can upload csrs as follows <https://docs.partner.android.com/gms/building/integrating/att-keys/rkp?authuser=1#upload-csrs>.

For example:

```
python device_info_uploader.py --credentials cred.json --json-csr csr.json --  
cache-token --company-id xxxxxx
```

Notice:

OEM can attach one csr json to another to achieve batch upload of csr requests.
For example `cat csr1.json >> csr.json`

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